

IDEA GENERATION

PROPOSED RESEARCH AGENDA FOR INTELLIGENT PRODUCT TRACEABILITY IN NIGERIA

PART REQUIREMENTS FOR SEMINAR

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MATRIC NO: [REDACTED]

AOI: INTELLIGENT PRODUCT TRACEABILITY

YEAR: 2025/2026

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E-TUTOR(S): [WITHHELD]

Research Problem: Intelligent Product Traceability and Blockchain Technology in Nigeria

Why does Nigeria urgently need empirically-tested blockchain traceability frameworks in its pharmaceutical and agri-food supply chains?

- The case for blockchain-based intelligent product traceability in Nigeria rests on a foundation of documented failure. Nigeria's pharmaceutical supply chain is one of the most compromised in sub-Saharan Africa, NAFDAC's own estimates place the proportion of substandard and counterfeit medicines in circulation at between 13% and 50%, depending on the category of product being assessed (Think Global Health, 2024). In agri-food, Nigeria recorded the highest number of food fraud incidents in sub-Saharan Africa in 2022 (Njiru *et al.*, 2025), with adulteration cases ranging from diluted beverages to outright product substitution at the wholesale level. These are not abstract regulatory concerns, they represent real, recurring failures of supply chain integrity that cost lives, erode consumer trust, and undermine the commercial viability of Nigerian manufacturers and exporters.
- Despite this urgent national reality, the academic and practical evidence base for blockchain-enabled product traceability in Nigeria remains remarkably thin. Akazue and colleagues (2023) developed a blockchain prototype for food value chain traceability in Nigeria, demonstrating technical feasibility in a laboratory setting, yet their work explicitly acknowledged the system had not been validated at scale, and no study since has built on their foundation. The global blockchain traceability literature, rich as it is, has overwhelmingly been designed for and validated in China, Europe, and North America, environments where internet connectivity is reliable, electricity is stable, and blockchain developers are plentiful. None of these conditions can be assumed in the Nigerian context, where broadband penetration stands at 48% nationally and where over 54% of the population remains offline (DataReportal, 2025). The scalability barriers identified by Cordeiro and Ferreira (2025), the smart contract cost concerns raised by Zhang and Cheng (2023), and the data source authenticity problem documented by Yao and Zhang (2022) all have amplified, contextually-specific dimensions in Nigeria that no existing study has examined.
- The problem is compounded by the regulatory environment. NAFDAC's Pharmaceutical Products Traceability Regulations, enacted on 30 October 2024, mandate unique identifiers and real-time data sharing across the entire pharmaceutical supply chain, from manufacturer to pharmacy, but do not prescribe the underlying technology (GS1 Nigeria, 2024). This creates an urgent practical question: can blockchain, with its well-documented advantages of immutability and decentralisation, be deployed in a scalable, interoperable, and cost-effective manner within Nigeria's specific infrastructure, regulatory, and institutional constraints? Existing literature provides no Nigerian-specific answer. This gap is not only academically significant, it is operationally pressing for thousands of Nigerian manufacturers, importers, distributors, and regulators who must make technology investment decisions under the new compliance mandate.
- The intellectual contribution this research proposes to make is a Nigeria-specific empirical framework that tests the relationship between blockchain technology's sub-dimensions smart contracts, decentralisation, scalability, data integrity, and interoperability, and the effectiveness of intelligent product traceability in Nigerian manufacturing supply chains. By generating local evidence rather than relying on extrapolations from foreign contexts, this study will provide actionable guidance for Nigerian businesses, NAFDAC, and technology providers navigating the mandatory traceability landscape. It will also contribute to the global literature by providing the first rigorous, context-grounded test of blockchain traceability theory in a major sub-Saharan African economy.

IDEA GENERATION · GAP 1 RESEARCH FRAMEWORK

Intelligent Product Traceability and Blockchain Technology in Nigerian Manufacturing

RESEARCH OBJECTIVE

To examine the extent to which blockchain technology sub-dimensions, smart contracts, decentralisation, scalability, data integrity, and interoperability, influence the effectiveness of intelligent product traceability in Nigerian manufacturing companies.

NULL HYPOTHESIS (H₀)

H₀: Smart contracts, decentralisation, scalability, data integrity, and interoperability of blockchain technology have no significant influence on the effectiveness of intelligent product traceability in Nigerian manufacturing companies.

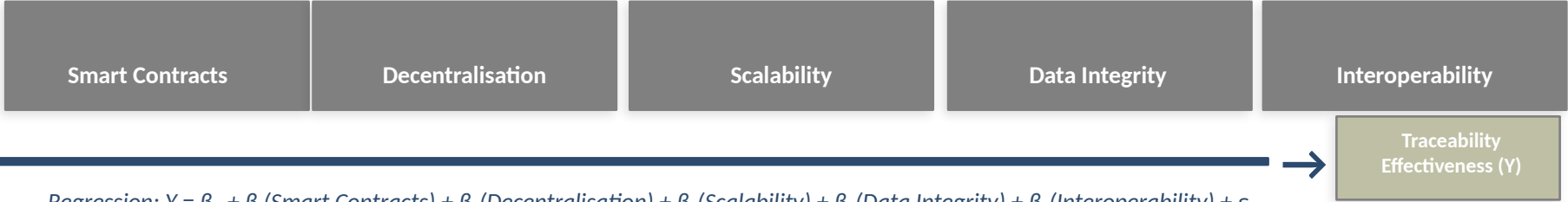
RESEARCH QUESTION

To what extent do smart contracts, decentralisation, scalability, data integrity, and interoperability of blockchain technology influence the effectiveness of intelligent product traceability in Nigerian manufacturing companies?

INTENDED CONTRIBUTION TO KNOWLEDGE

This study will generate the first empirically-tested, Nigeria-specific model of blockchain-enabled product traceability, closing a critical gap in both African supply chain management literature and the global blockchain traceability evidence base, while providing actionable guidance for NAFDAC-compliant technology adoption.

OPERATIONALIZATION OF VARIABLES



Research Problem: Intelligent Product Traceability and IoT Integration in Nigeria

Can IoT-enabled traceability systems be effectively designed and deployed in Nigeria's low-connectivity, resource-constrained supply chain environments?

- One of the most fundamental and underappreciated problems in designing intelligent product traceability systems for Nigeria is the connectivity assumption. The global IoT traceability literature, the studies that inform current understanding of how RFID tags, GPS sensors, temperature monitors, and QR code readers contribute to supply chain visibility, almost universally assumes a digital environment that simply does not exist across much of Nigeria's agricultural and manufacturing heartland. As of January 2025, more than 54% of Nigeria's 220 million people were not connected to the internet (DataReportal, 2025). Power supply failures are endemic, with businesses in cities like Lagos, Kano, and Port Harcourt relying on private diesel generators for significant portions of their operational hours. And the cost of deploying sensor networks, a dimension that Thakur, Ahuja and Singh (2024) explicitly flag as under-investigated even in high-income country contexts, is proportionally far higher in Nigeria's small and medium enterprise sector, which constitutes the backbone of the country's manufacturing and agri-food industries.
- This matters enormously for product traceability. The physical-digital interface, the point where a physical product receives and maintains its digital identity through RFID tags, QR codes, or embedded sensors, is only as reliable as the infrastructure that supports it. Huang, Xie, Ning and Hu (2021) found that even basic QR code recognition can fail under poor lighting conditions; in Nigerian open-air markets and rural processing environments, such conditions are routine rather than exceptional. The oracle problem identified by Tsang *et al.*, (2019), the challenge of ensuring that physical product data is not tampered with before it enters the digital ledger, is acutely relevant in contexts where informal market actors with limited technology literacy are handling products at the distribution stage. Dib, Rahmouni and colleagues (2023) demonstrated the potential of IoT-enabled live data transmission for supply chain traceability, but their system required continuous GSM connectivity, a requirement that renders it impractical across large parts of Nigeria without significant adaptation.
- The research opportunity here is specific, practically urgent, and almost entirely unoccupied in the literature. No published study has examined how IoT integration sub-variables, real-time sensing, RFID and QR tracking, edge computing, data security, and connectivity, influence the effectiveness of intelligent product traceability within the specific operational constraints of Nigeria's manufacturing and agri-food sectors. The government's Fiber Forward project (90,000 km of new fibre optic infrastructure) and the 774 LGA Connectivity initiative are beginning to shift the connectivity landscape (U.S. ITA, 2024), but these investments will take years to fully materialise. In the meantime, traceability system designers and Nigerian businesses need evidence-based guidance on which IoT components deliver the greatest traceability value per naira of investment in Nigeria's specific infrastructure environment and current literature provides none.
- The contribution this research makes is twofold. First, it generates empirical evidence on the IoT-traceability relationship from an African emerging economy perspective, filling a specific geographical and contextual gap that existing Chinese-, European-, and American-validated models cannot fill. Second, by disaggregating IoT integration into its component sub-dimensions and testing each against traceability effectiveness, this study will provide practical, prioritised guidance to Nigerian supply chain managers on where to invest first, an operationally valuable finding that generic technology advocacy literature does not offer.

Intelligent Product Traceability and IoT Integration in Nigerian Manufacturing

RESEARCH OBJECTIVE

To investigate the influence of IoT integration sub-dimensions, real-time sensing, RFID/QR tracking, edge computing, data security, and connectivity, on the effectiveness of intelligent product traceability in Nigerian manufacturing companies.

NULL HYPOTHESIS (H₀)

H₀: Real-time sensing, RFID/QR tracking, edge computing, data security, and connectivity dimensions of IoT integration have no significant influence on the effectiveness of intelligent product traceability in Nigerian manufacturing companies.

RESEARCH QUESTION

In what ways do real-time sensing, RFID/QR tracking, edge computing, data security, and connectivity dimensions of IoT integration influence the effectiveness of intelligent product traceability in Nigerian manufacturing companies?

INTENDED CONTRIBUTION TO KNOWLEDGE

This study will produce the first IoT-traceability effectiveness model validated in a sub-Saharan African manufacturing context, providing a contextually-grounded alternative to models derived exclusively from high-income country environments, and offering practical investment prioritisation guidance for Nigerian supply chain managers.

OPERATIONALIZATION OF VARIABLES



Regression: $Y = \beta_0 + \beta_1(\text{Real-Time Sensing}) + \beta_2(\text{RFID/QR Tracking}) + \beta_3(\text{Edge Computing}) + \beta_4(\text{Data Security}) + \beta_5(\text{Connectivity}) + \varepsilon$

Research Problem: Intelligent Product Traceability and Supply Chain Transparency in Nigeria

Does intelligent product traceability translate into genuine supply chain transparency in Nigeria's fragmented, multi-stakeholder distribution environment?

- There is a deceptively important distinction at the heart of this research gap, one that the broader traceability literature has only recently begun to acknowledge. Traceability and transparency are not the same thing. Traceability is a technical capacity: the ability to identify, track, and document the movement and attributes of a product at every stage of its supply chain journey. Transparency is a social and organisational achievement: the willingness and ability of supply chain actors to share that traceability data openly, accurately, and in ways that produce accountability. Garcia-Torres, Rey-Garcia and Saenz (2024) make this distinction with theoretical precision, demonstrating that stakeholder collaboration is a necessary precursor to both constructs and that without deliberate governance investment, even the most sophisticated traceability system can fail to produce the transparency it theoretically enables.
- Nigeria's supply chain structure makes this distinction particularly consequential. The country's distribution networks are characterised by deep informality, multiple-layer middlemen systems, and institutional trust deficits that make data sharing across organisational boundaries genuinely difficult. When a product changes hands between a manufacturer, a national distributor, a regional wholesaler, a state-level sub-distributor, and an open-market retailer, as is common across Nigeria's fast-moving consumer goods and food sectors, each handover represents a potential point of information loss, deliberate omission, or active data falsification. Langton, Mafini and Tlale (2026) found in their South African manufacturing study that immutability and provenance practices drove supply chain transparency, but that security features alone did not, suggesting that the social dimensions of transparency are as important as the technical ones. This finding has never been tested in Nigeria, where the institutional context for data sharing is even more fragmented than in South Africa.
- Nigeria's designation as the AfCFTA Digital Trade Champion in 2024 adds a further dimension of urgency. The AfCFTA Digital Trade Protocol, adopted in February 2024, establishes legal frameworks for cross-border digital supply chain data sharing across Africa. For Nigerian exporters to benefit from these frameworks, they will need not only the technical capacity to generate traceability data but also the organisational practices and stakeholder relationships to share that data credibly with trading partners across the continent. Khan, Alshahrani and Jacquemod (2023) found empirically that digital platforms alone do not significantly improve information quality without the mediation of organisational transparency practices, a finding that points directly to the governance investment Nigeria's export-oriented manufacturers will need to make alongside any technology deployment. Yet no study has examined this relationship in Nigeria, where the regulatory mandate (NAFDAC Traceability Regulations 2024, NSPRI Food Safety Policy 2024–2030) is creating new transparency obligations that Nigerian companies are only beginning to understand and operationalise.
- The proposed research addresses this gap by examining how provenance verification, information sharing, stakeholder collaboration, supply chain visibility, and fraud reduction practices, the five sub-dimensions of supply chain transparency identified from the literature, translate into effective intelligent product traceability outcomes in Nigerian manufacturing. The findings will contribute to theory by providing the first empirical test of the traceability-transparency relationship in a West African context, and to practice by identifying which transparency practices generate the greatest traceability returns in Nigeria's specific institutional environment.

IDEA GENERATION · GAP 3 RESEARCH FRAMEWORK

Intelligent Product Traceability and Supply Chain Transparency in Nigerian Manufacturing

RESEARCH OBJECTIVE

To examine how provenance verification, information sharing, stakeholder collaboration, supply chain visibility, and fraud reduction practices influence the effectiveness of intelligent product traceability in Nigerian manufacturing companies.

NULL HYPOTHESIS (H₀)

H₀: Provenance verification, information sharing, stakeholder collaboration, supply chain visibility, and fraud reduction practices have no significant influence on the effectiveness of intelligent product traceability in Nigerian manufacturing companies.

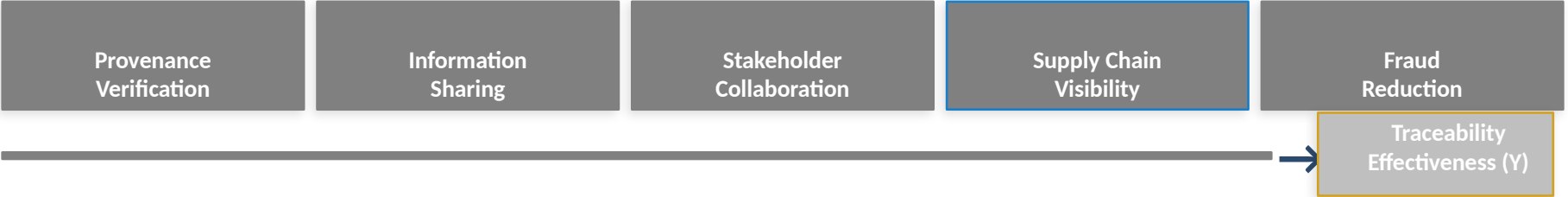
RESEARCH QUESTION

To what extent do provenance verification, information sharing, stakeholder collaboration, supply chain visibility, and fraud reduction practices influence the effectiveness of intelligent product traceability in Nigerian manufacturing companies?

INTENDED CONTRIBUTION TO KNOWLEDGE

This research will deliver the first empirical test of the traceability-transparency nexus in West Africa, extending Garcia-Torres et al.'s (2024) theoretical framework to an emerging economy context and providing evidence-based guidance for Nigerian companies navigating new regulatory transparency mandates under NAFDAC 2024 and the AfCFTA Digital Trade Protocol.

OPERATIONALIZATION OF VARIABLES



Regression: $Y = \beta_0 + \beta_1(\text{Provenance Verification}) + \beta_2(\text{Information Sharing}) + \beta_3(\text{Stakeholder Collaboration}) + \beta_4(\text{SC Visibility}) + \beta_5(\text{Fraud Reduction}) + \epsilon$

Research Problem: Intelligent Product Traceability and Consumer Trust / Food Safety in Nigeria

Do Nigerian consumers trust traceability information and does that trust influence their behaviour and willingness to pay for traceable products?

- Nigeria has a documented product safety crisis that is, at its core, a crisis of consumer trust. When NAFDAC acknowledges that a significant fraction of medicines in circulation are counterfeit, when Nigerian consumers survey after survey report awareness of 'plastic rice,' fake dairy products, and adulterated beverages, the question for intelligent product traceability research is not simply whether traceability systems can be built, but whether consumers will trust them, understand them, use them to make decisions, and pay a premium for the safety they theoretically guarantee. These are fundamentally behavioural and psychological questions, and they are almost entirely unaddressed in the Nigerian academic literature.
- The global evidence base here is instructive but geographically limited. Liu, Yu *et al.*, (2024) demonstrated that traceability label trust significantly influences Chinese consumers' purchasing decisions for traceable pork, with food safety identification strengthening the relationship. Nguyen and Nguyen (2025) showed that blockchain-enabled reviews and QR code traceability together drive Vietnamese consumers' willingness to pay premium prices for sustainably traced products, but simultaneously acknowledged that no validated multidimensional scale yet exists for measuring traceability-based consumer trust. Maesano, Sadrmousavigargari and Castellini (2025) found that while Italian consumers recognise blockchain's transparency potential, they do not yet fully understand or act on it. And Tseng, *et al.*, (2022) showed that their China-focused TAM-IS traceability acceptance model achieved 78.3% explanatory power, but called explicitly for replication in other national contexts.
- Nigeria represents perhaps the most important national context in which consumer traceability behaviour has not yet been studied. The stakes are higher, counterfeit products cause demonstrable harm. The trust baseline is lower, years of exposure to substandard goods have left Nigerian consumers deeply sceptical of product claims they cannot independently verify. The technological landscape is different, with 150 million mobile connections and QR code scanning increasingly accessible even to lower-income consumers, the technical infrastructure for consumer-facing traceability exists, but the behavioural response to it remains unknown. Charan and Reddy (2026) identify that technological complexity and high costs constrain adoption, particularly for smaller producers, a finding with direct implications for Nigeria's vast informal and semi-formal food production sector, where the producers most in need of traceability credentials are often the least able to afford them.
- What this research proposes is the first Nigeria-grounded empirical study of how traceability label trust, purchase behaviour, technology acceptance, safety verification, and willingness to pay are interrelated, and how these consumer-side variables are shaped by intelligent product traceability system design. The findings will not only contribute to consumer behaviour theory in an African emerging economy context, they will provide practical design guidance for traceability system developers and Nigerian food safety regulators on what consumers actually need to see, understand, and experience before traceability investments translate into market confidence and commercial returns.

Intelligent Product Traceability and Consumer Trust / Food Safety in Nigerian Manufacturing

RESEARCH OBJECTIVE

To assess how traceability label trust, purchase behaviour, technology acceptance, safety verification, and willingness to pay influence the effectiveness of intelligent product traceability as a consumer-facing mechanism in Nigeria's food and pharmaceutical supply chains.

RESEARCH QUESTION

To what extent do traceability label trust, purchase behaviour, technology acceptance, safety verification, and willingness to pay influence the effectiveness of intelligent product traceability in Nigeria's food and pharmaceutical supply chains?

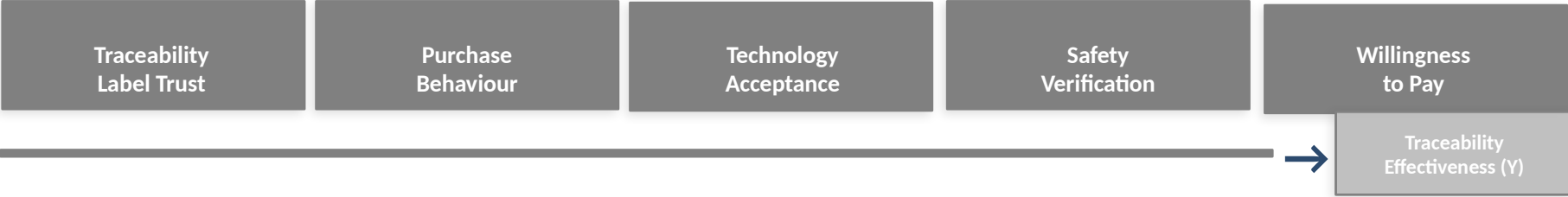
NULL HYPOTHESIS (H₀)

H₀: Traceability label trust, purchase behaviour, technology acceptance, safety verification, and willingness to pay have no significant influence on the effectiveness of intelligent product traceability as a consumer-facing safety mechanism in Nigeria's food and pharmaceutical supply chains.

INTENDED CONTRIBUTION TO KNOWLEDGE

This study will generate the first validated consumer traceability behaviour model for Nigeria and sub-Saharan Africa , filling the measurement gap identified by Nguyen & Nguyen (2025) and extending Liu et al.'s (2024) Chinese model to an African context, while providing actionable guidance for NAFDAC's public engagement strategy under the 2024 Traceability Regulations.

OPERATIONALIZATION OF VARIABLES



Regression: $Y = \beta_0 + \beta_1(\text{Label Trust}) + \beta_2(\text{Purchase Behaviour}) + \beta_3(\text{Technology Acceptance}) + \beta_4(\text{Safety Verification}) + \beta_5(\text{Willingness to Pay}) + \epsilon$

Research Problem: Intelligent Product Traceability and Organisational Performance / Sustainability in Nigeria

Does investing in intelligent product traceability generate measurable performance and sustainability returns for Nigerian manufacturers and can this be empirically proven?

- The business case for intelligent product traceability in Nigeria is, for most manufacturing companies, still made on the basis of compliance, the cost of not doing it is now higher than the cost of doing it, given NAFDAC's 2024 regulatory mandates and the reputational damage of documented product safety failures. But compliance-driven investment rarely generates strategic advantage. The more important and intellectually richer question, and the one this research seeks to answer, is whether intelligent product traceability, beyond meeting regulatory requirements, actually generates measurable improvements in operational efficiency, organisational sustainability, counterfeit reduction, supply chain integrity, and competitive advantage for Nigerian manufacturing firms.
- Globaly Khan *et al.*, (2022) establish empirically that traceability technology, mediated by supply chain transparency, has a significant positive impact on organisational performance. Abobassier *et al.*, (2026) demonstrate in Turkish manufacturing SMEs that blockchain-enabled traceability positively influences sustainable supply chain practices, both directly and through the mediation of information transparency and stakeholder collaboration. Onu, Mbohwa and Pradhan (2024) frame blockchain-powered traceability as a direct instrument for eliminating counterfeit products and strengthening supply chain integrity across industries, a strategic outcome with immediate relevance to Nigeria, where the infiltration of substandard goods is an acknowledged multi-sector crisis. Yet none of these findings has been tested in Nigeria. The Turkish SME result, compelling as it is, was derived in a context with fundamentally different institutional infrastructure, regulatory environment, and supply chain structure from Nigeria's. The claim that traceability drives performance and sustainability in Nigerian manufacturers cannot be assumed from foreign evidence, it must be demonstrated through local research.
- Nigeria's agricultural sector offers an especially compelling case for the sustainability dimension of this argument. Post-harvest losses across Nigerian agricultural supply chains are estimated at 30% to 50% annually, billions of dollars of wasted food, squandered inputs, and unnecessary carbon emissions that intelligent traceability systems could materially reduce by enabling faster anomaly detection, more precise cold chain management, and earlier intervention when product condition deteriorates. These losses represent not just an economic failure but a sustainability crisis directly relevant to SDG 2 (Zero Hunger), SDG 12 (Responsible Consumption and Production), and SDG 13 (Climate Action), goals that the NSPRI Food Safety Policy (2024–2030) explicitly references as strategic priorities. Yet no published study has quantified what sustainability return Nigerian manufacturers can expect from investing in intelligent traceability systems, making evidence-based business case development practically impossible for private sector decision-makers.
- The final dimension competitive advantage, is arguably the most strategically significant in the medium term. As Nigeria's AfCFTA Digital Trade Champion status creates new obligations and opportunities for Nigerian exporters, the ability to demonstrate verifiable product authenticity, origin, and handling standards through intelligent traceability systems will increasingly become a prerequisite for market access across the continent. Companies that invest now will be positioned as preferred suppliers; those that delay will face growing market exclusion. This study proposes to be the first to test whether operational efficiency, sustainability practices, counterfeit reduction, supply chain integrity, and competitive advantage, the five operationalised sub-dimensions of organisational performance, are significantly driven by intelligent product traceability investment in Nigerian manufacturing, providing the empirical foundation for a compelling, evidence-based strategic argument for traceability adoption that compliance mandates alone cannot deliver.

Intelligent Product Traceability and Organisational Performance / Sustainability in Nigerian Manufacturing

RESEARCH OBJECTIVE

To investigate the influence of intelligent product traceability on operational efficiency, sustainability practices, counterfeit reduction, supply chain integrity, and competitive advantage in Nigerian manufacturing companies.

NULL HYPOTHESIS (H₀)

H₀: Intelligent product traceability has no significant influence on operational efficiency, sustainability practices, counterfeit reduction, supply chain integrity, and competitive advantage in Nigerian manufacturing companies.

RESEARCH QUESTION

To what extent does intelligent product traceability influence operational efficiency, sustainability practices, counterfeit reduction, supply chain integrity, and competitive advantage in Nigerian manufacturing companies?

INTENDED CONTRIBUTION TO KNOWLEDGE

This study will generate the first Nigerian empirical evidence linking intelligent traceability investment to measurable organisational performance and sustainability outcomes, providing a quantified strategic business case for traceability adoption that extends beyond regulatory compliance and directly supports Nigeria's SDG and AfCFTA export competitiveness commitments.

OPERATIONALIZATION OF VARIABLES

Operational
Efficiency

Sustainability
Practices

Counterfeit
Reduction

Supply Chain
Integrity

Competitive
Advantage

Organisational
Performance (Y)

Regression: $Y = \beta_0 + \beta_1(\text{Operational Efficiency}) + \beta_2(\text{Sustainability}) + \beta_3(\text{Counterfeit Reduction}) + \beta_4(\text{SC Integrity}) + \beta_5(\text{Competitive Advantage}) + \varepsilon$

IDEA GENERATION • CONCEPTUAL MODEL

Integrated Conceptual Framework: Intelligent Product Traceability and Organisational Outcomes in Nigeria



GAP 5: Traceability Effectiveness (Y) → Organisational Performance (Operational Efficiency | Sustainability | Counterfeit Reduction | SC Integrity | Competitive Advantage)

IDEA GENERATION · PROPOSED RESEARCH TOPICS

Three Research Topic Proposals Derived from the Five Gaps

01

Blockchain and IoT Integration for Intelligent Product Traceability: An Empirical Study of Nigerian Pharmaceutical and Agri-Food Manufacturing Companies

Integrates Gaps 1 and 2. Tests the combined influence of blockchain technology sub-dimensions and IoT integration components on traceability effectiveness, with a focus on Nigeria's NAFDAC-regulated pharmaceutical and agri-food manufacturing sectors. Addresses the scalability, connectivity, and data integrity challenges that are uniquely amplified in Nigeria's infrastructure context.

02

Supply Chain Transparency, Consumer Trust, and Intelligent Product Traceability Effectiveness: Evidence from Nigerian Manufacturing Firms

Integrates Gaps 3 and 4. Examines the dual-pathway through which intelligent traceability generates value, the organisational transparency pathway (provenance verification, information sharing, stakeholder collaboration) and the consumer trust pathway (label trust, safety verification, willingness to pay) and tests their relative influence on traceability effectiveness in Nigerian manufacturers.

03

Intelligent Product Traceability as a Strategic Resource: Empirical Evidence on Organisational Performance and Sustainability in Nigerian Manufacturing

Focuses on Gap 5. Applies the Resource-Based View to test whether intelligent product traceability investment generates measurable competitive advantage, operational efficiency gains, counterfeit reduction, and sustainability returns for Nigerian manufacturing companies — providing the quantified strategic business case that compliance-driven investment arguments cannot supply.